

Number: P3375R3  
Date: 2025-05-12 (16:30)  
Audience: SG6, SG14, LEWG  
Reply-to: [Guy Davidson](#)

## Table of contents

- [Abstract](#)
- [Revision history](#)
- [Motivation](#)
- [Status quo](#)
- [Prior art](#)
- [Possible approaches](#)
- [Related papers](#)
- [Discussion](#)
- [Open questions](#)
- [References and bibliography](#)

## Contributors

Many thanks to the following people who have helped to bring this proposal to life:

- Andrew Drakeford
- David Olsen
- Hana Dušíková
- Hans Boehm
- Jens Maurer
- Joshua Cranmer
- Oliver Rosten
- Robin Leroy
- Sam Saariste
- Sherry Ignatchenko
- Shreyas Atre
- Simon Byrne
- Tim Mattson
- Tom Deakin
- Walter E. Brown

This proposal was made possible with the sponsorship of [Six Impossible Things Before Breakfast](#).

# Reproducible floating-point results

## 1. Abstract

The C++ standard, hereafter referred to as [\[C++\]](#), has very little to say about floating-point specification, which means that it is extremely hard to write floating-point expressions that, given the same inputs and rounding mode, will produce the same results on all implementations. This paper proposes the introduction of a new set of functions which specifies sufficient conformance with the provisions specified in ISO/IEC 60559:2020 (which is itself imported from IEEE Std 754-2019, unmodified except for editorial adjustment), hereafter referred to as [\[60559\]](#), to guarantee this reproducibility.

## 2. Revision history

### R3 2025-05-12

The desired outcome of this revision is for SG6 to decide on an approach to achieve reproducibility and to forward this proposal to LEWG.

- Extended the contributor list. Many thanks, all of you.
- Updated the abstract.
- Updated the banking discussion.
- Added discussion of the Technical Specification regarding floating-point extensions for C.
- Added correctly rounded functions to the list of possible approaches.
- Changed discussion to focus on correctly rounded functions, and to reduce scope creep.
- Updated open questions.
- Updated the bibliography.

### R2 2025-01-13